Ethanol Blending in Gasoline - Mexico

August 2025

U.S. GRAINS & BIOPRODUCTS COUNCIL

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Ethanol Blending in Latin America

There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Latin America to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.

Sixteen countries with potential and additional use of ethanol were studied: 1) gasoline market profiles; 2) Optimization of gasoline blends with ethanol and 3) Environmental impact of gasolines blended with ethanol.

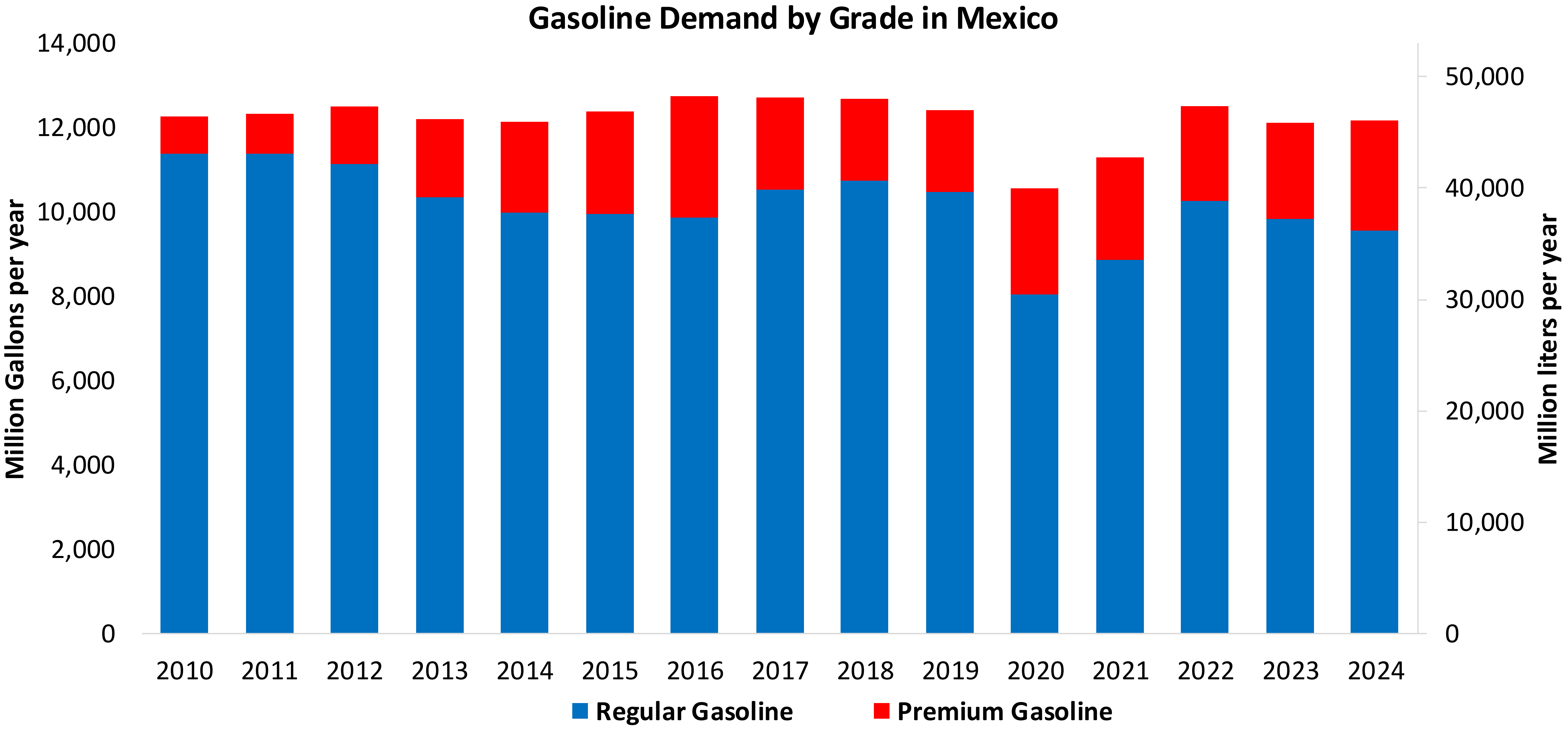
Ethanol Blending in Gasoline - Mexico

In 2024, gasoline consumption was 12,200 million gallons (46,000 million liters). There are two grades of gasoline, Regular (AKI 87) and Premium (AKI 91). Regular gasoline is the main grade in the market, representing 90% of the volume. There are stringent gasoline specifications for major metropolitan areas in Mexico City, Guadalajara and Monterrey. Demand is much higher than production, making necessary to import 60% of gasoline mainly from the United States.

Gasoline quality norm (NOM-016-CRE de 2016) allows blends with a maximum of 5.8% v/v of ethanol with the exception of metropolitan areas of Valle de Mexico, Guadalajara and Monterrey. Current ethanol use is equivalent to an average of 1.06% in the Rest of the Country grade gasoline. In 2025, a new biofuels law was approved that fosters the production with agricultural and other organic residuals. Recently there are new state level efforts to increase ethanol consumption (e.g. Tamaulipas).

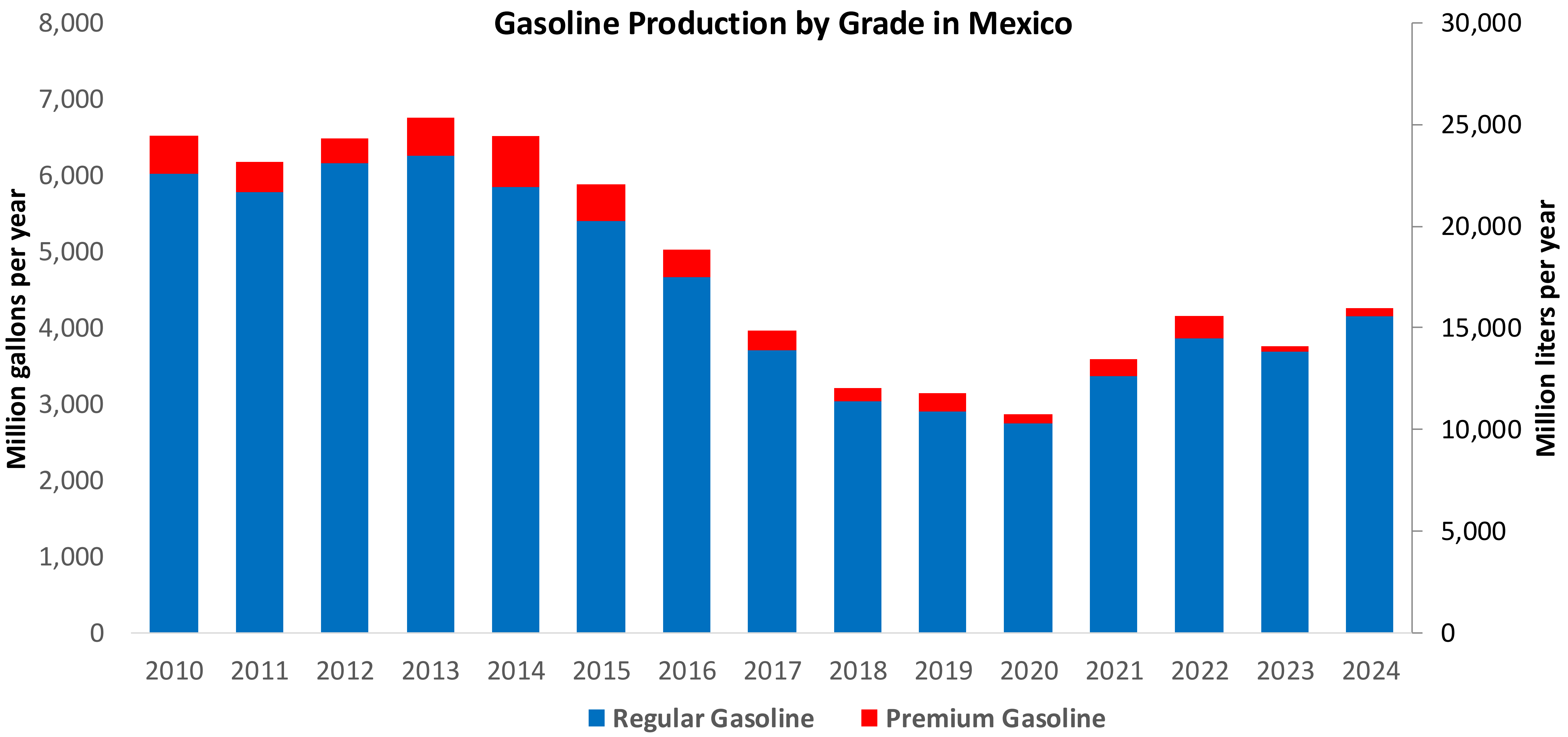
Source: SIE - SENER, 2024

Gasoline Demand in Mexico



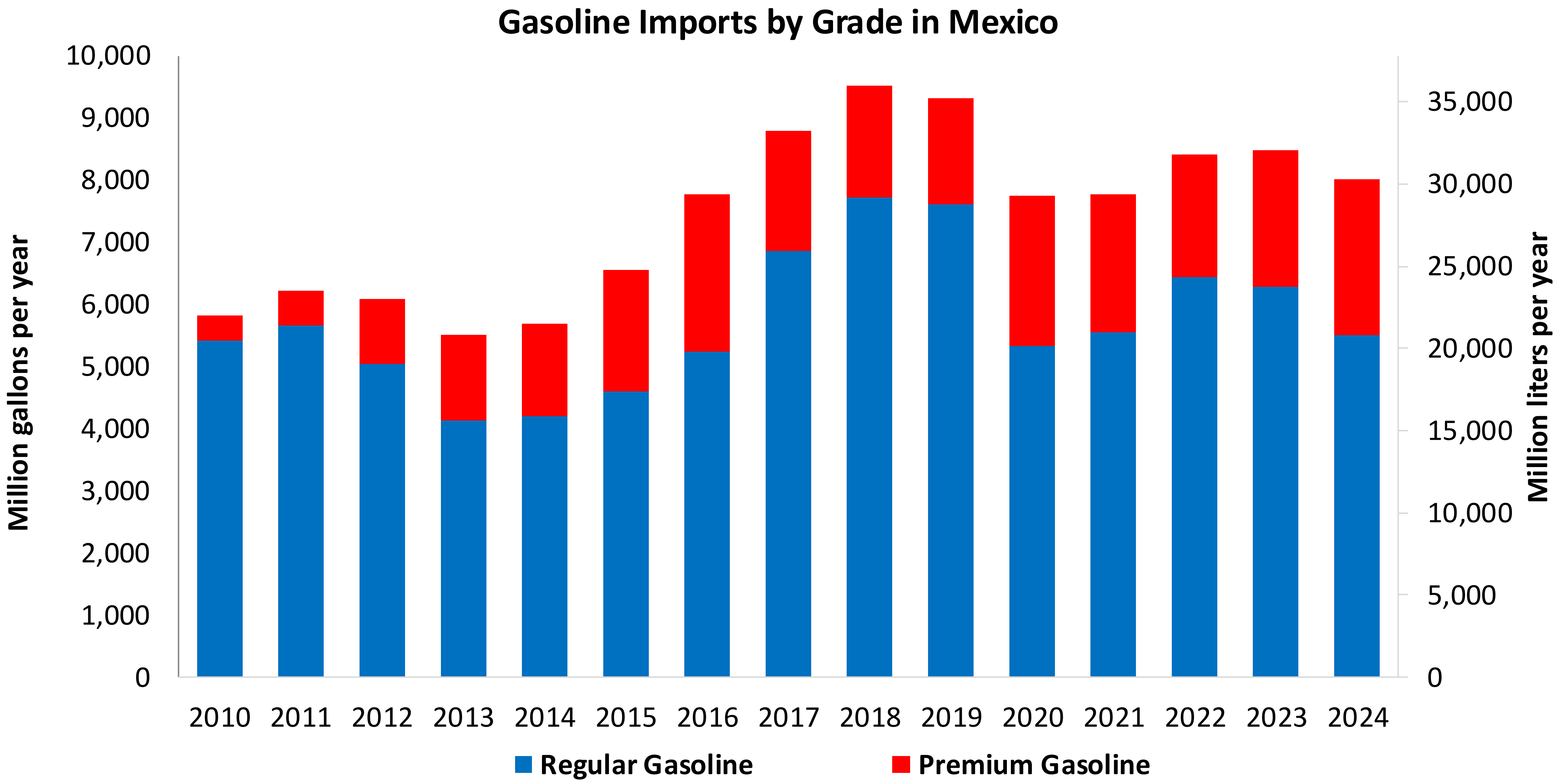
Source: SIE - SENER, 2025

Gasoline Production in Mexico



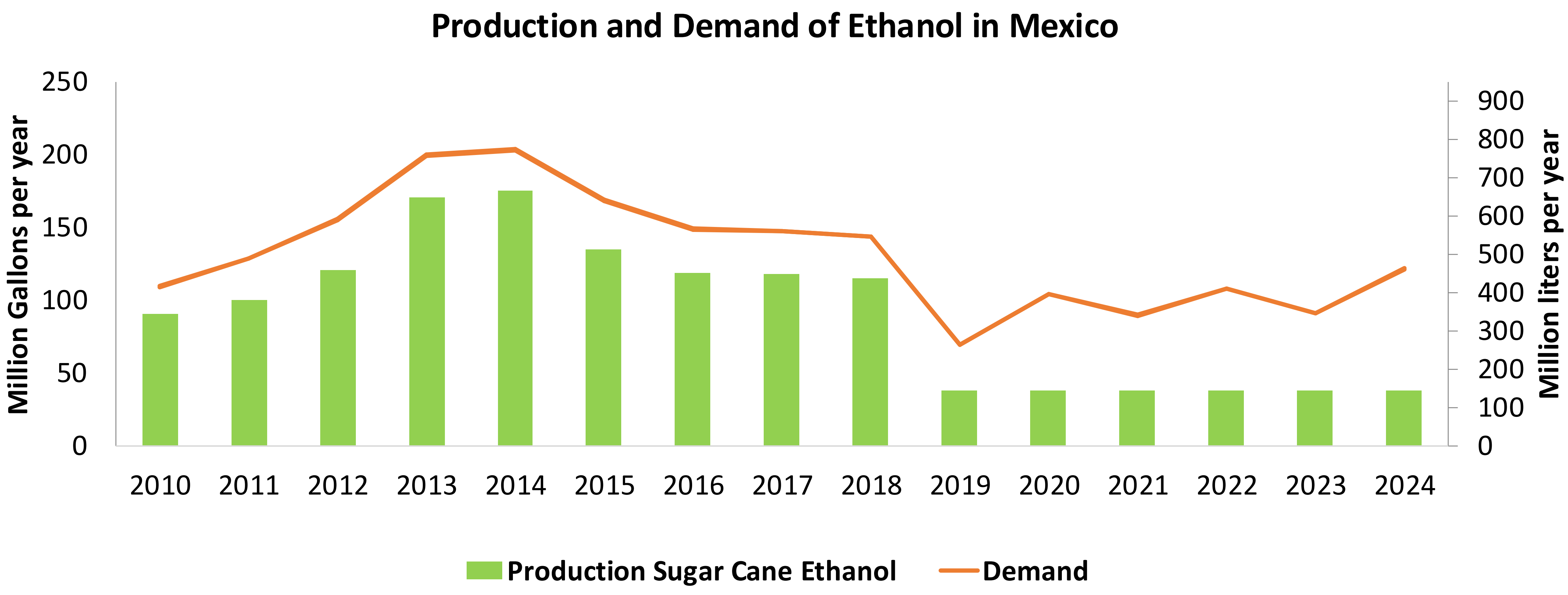
Source: SIE - SENER, 2025

Gasoline Imports in Mexico



Source: SIE - SENER, 2025

Ethanol Balance in Mexico



Source: CEDRSSA, 2020; EIA, 2025; Sanchez, 2014.

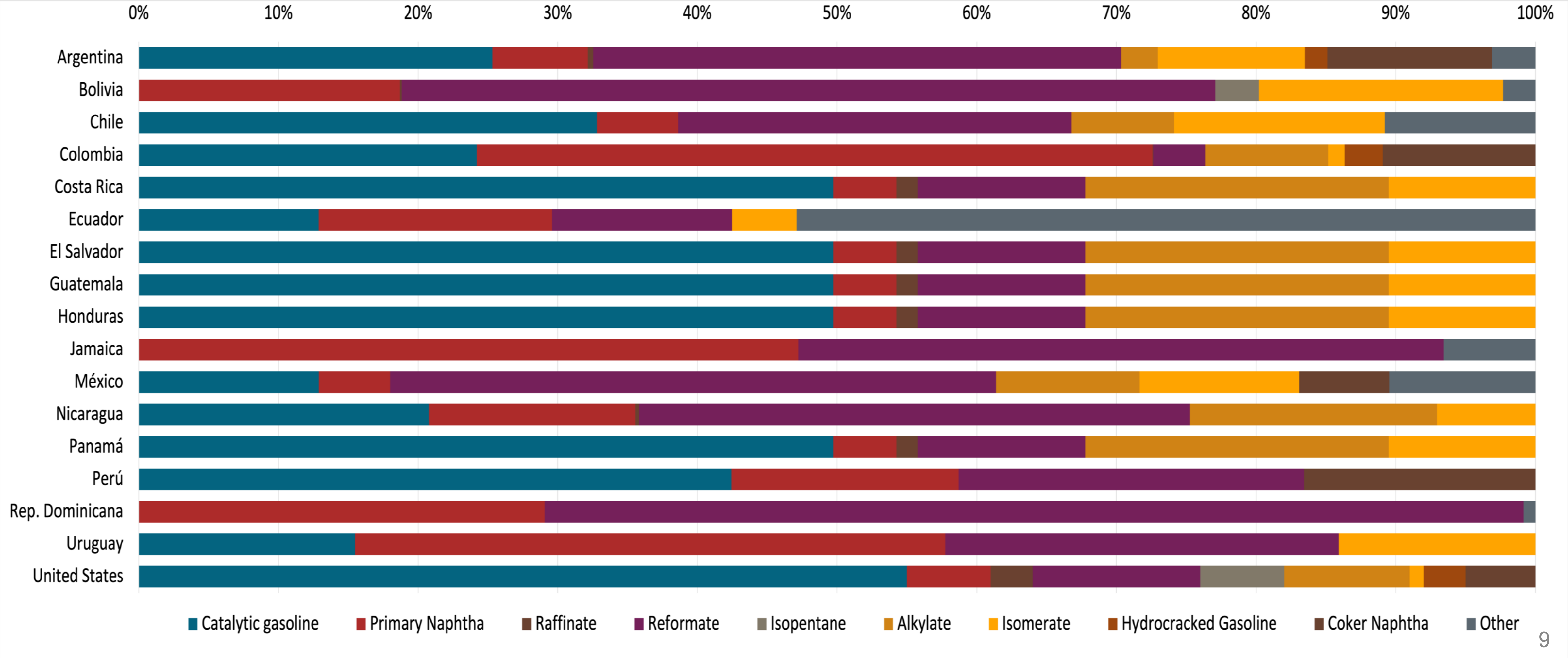
Gasoline Quality in Mexico

Name	NOM-016-CRE-2016						EN 228:2012 + A1:2017 (Euro 6 enabling)			
Implementation Date	2016						2017			
Applicability	Mexico City	Mexico City	Guadalajara and Monterrey	Guadalajara and Monterrey	Rest of the country	Rest of the country	All countries			
Selected Grade	Gasoline Regular	Gasoline Premium	Gasoline Regular	Gasoline Premium	Gasoline Regular	Gasoline Premium	RON 95 E5	RON 95 E10	RON 98 E5	RON 98 E10
Benzene Content	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 2 %v/v	< 2 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v
Aromatics	< 25 %v/v	< 25 %v/v	< 32 %v/v	22,6% v/v	Report	22,6% v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v
Olefins	< 10 %v/v	< 10 %v/v	< 11,9 %v/v	< 11,9 %v/v	Report	< 2,5 mg/l	< 18 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v
Lead Content	-	-	-	-	-	-	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l
Manganese	0 mg/l	0 mg/l	0 mg/l	0 mg/l	0 mg/l	0 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l
RON		94		94		94	> 95	> 95	> 98	> 98
MON	82		82		82		> 85	> 88	> 85	> 88
AKI										
Sulfur Content	< 80 mg/kg	< 80 mg/kg	< 80 mg/kg	< 80 mg/kg	< 80 mg/kg	< 80 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg
Oxygen Content	< 3,7 %m/m	< 2,7 %m/m	< 3,7 %m/m	< 2,7 %m/m	< 3,7 %m/m	< 2,7 %m/m	<2,7 % m/m	<3,7 % m/m	<2,7 % m/m	<3,7 % m/m
Ethanol (EtOH)	Not allowed	Not allowed	Not allowed	Not allowed	> 5,8 %v/v	> 5,8 %v/v	<5 %v/v	<10 %v/v	<5 %v/v	<10 %v/v
RVP 37.8°C (Summer)	< 62 kPa	< 62 kPa	< 62 kPa	< 62 kPa	< 69 kPa North, < 62 kPa Center, South, Pacific	< 69 kPa North, < 62 kPa Center, South, Pacific	<> 60 - 70 kPa *Depends on the country, RVP is regulated in the EU Fuel Quality Directive			
RVP 37.8 °C(Winter)	< 79 kPa	< 79 kPa	< 79 kPa	< 79 kPa	< 79 kPa	< 79 kPa				
RVP 37.8°C (Transition)	< 54 kPa	< 54 kPa	< 54 kPa Guadalajara	< 54 kPa Guadalajara						
MTBE							-	-	-	-
Ethers 5 or more C Atoms	-	-	-	-	-	-	Based on oxygen content	<22 %v/v	Based on oxygen content	<22 %v/v

Source: CRE, 2016

Gasoline Component Blending in Latin America

Gasoline is a blend of a base gasoline and other components. This blending is usually done at blending terminals as only 30% of the world's finished gasoline is distributed directly from refineries. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Latin America are:



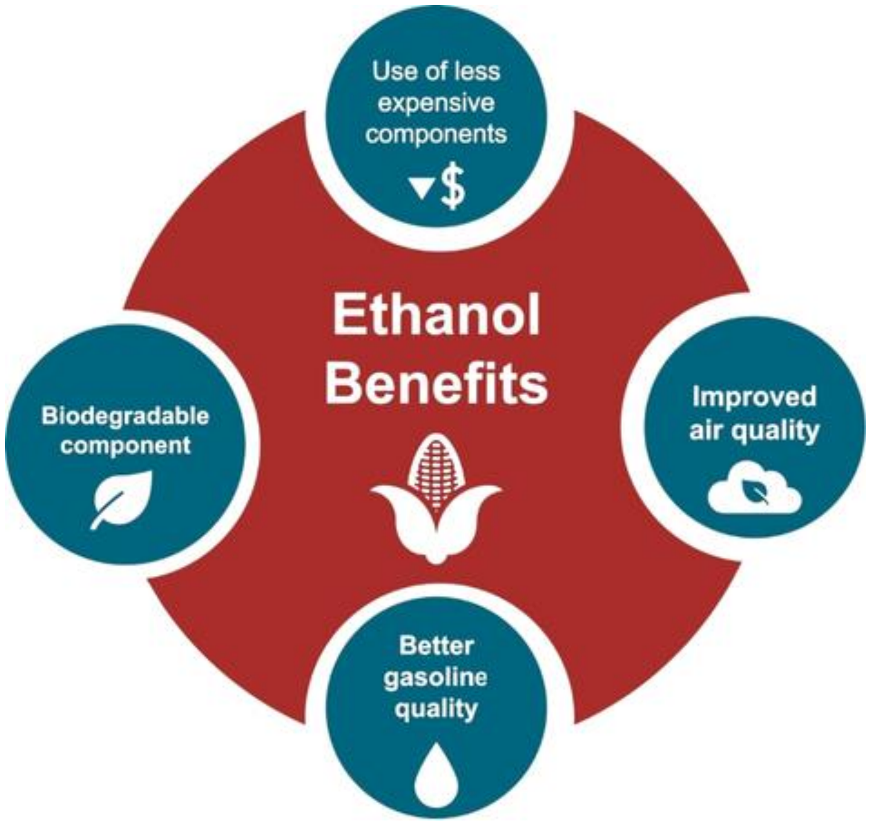
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component.The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives.To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

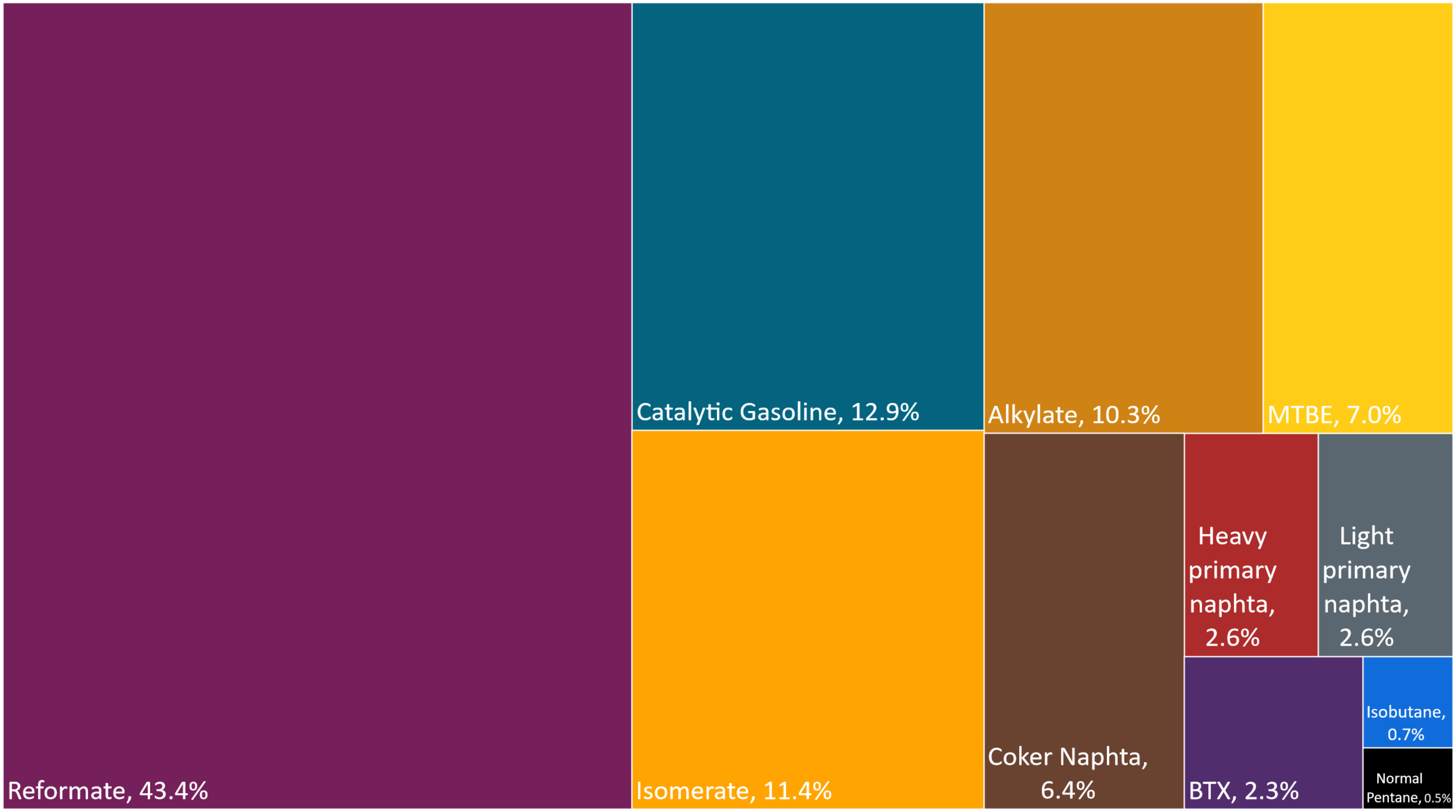
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

The blending model uses gasoline component spot average prices 2024 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.

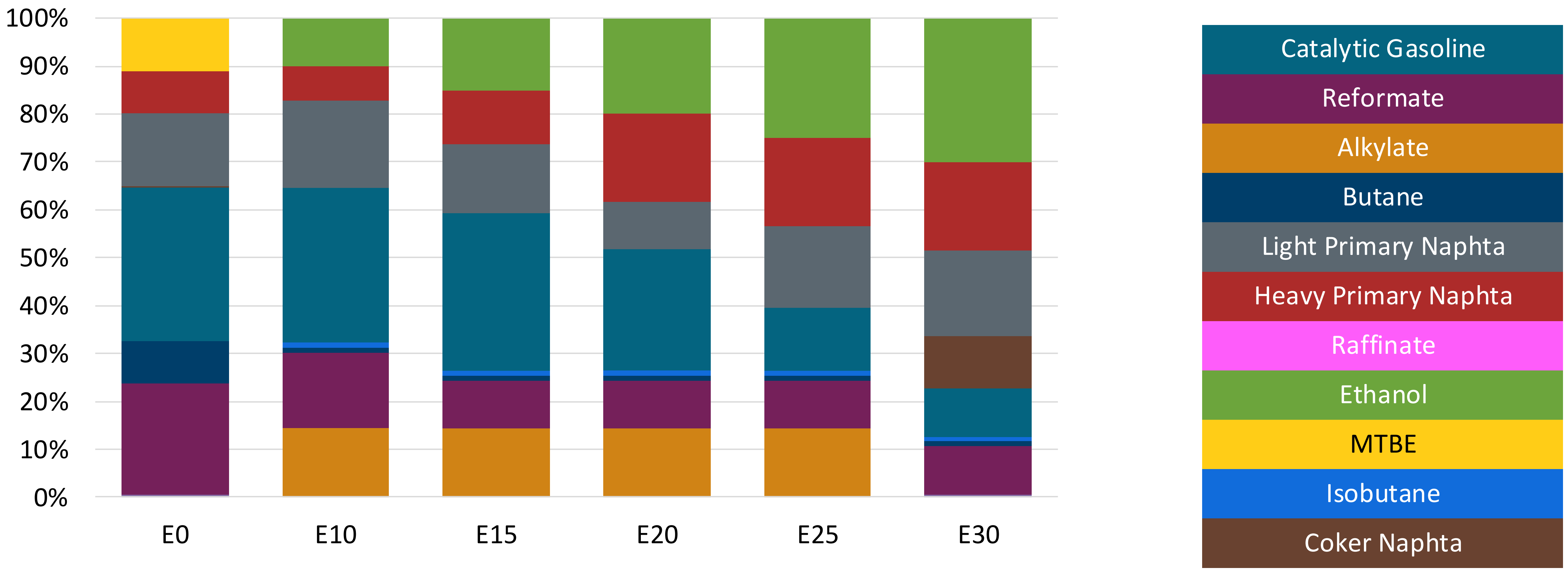


Available Blending Components



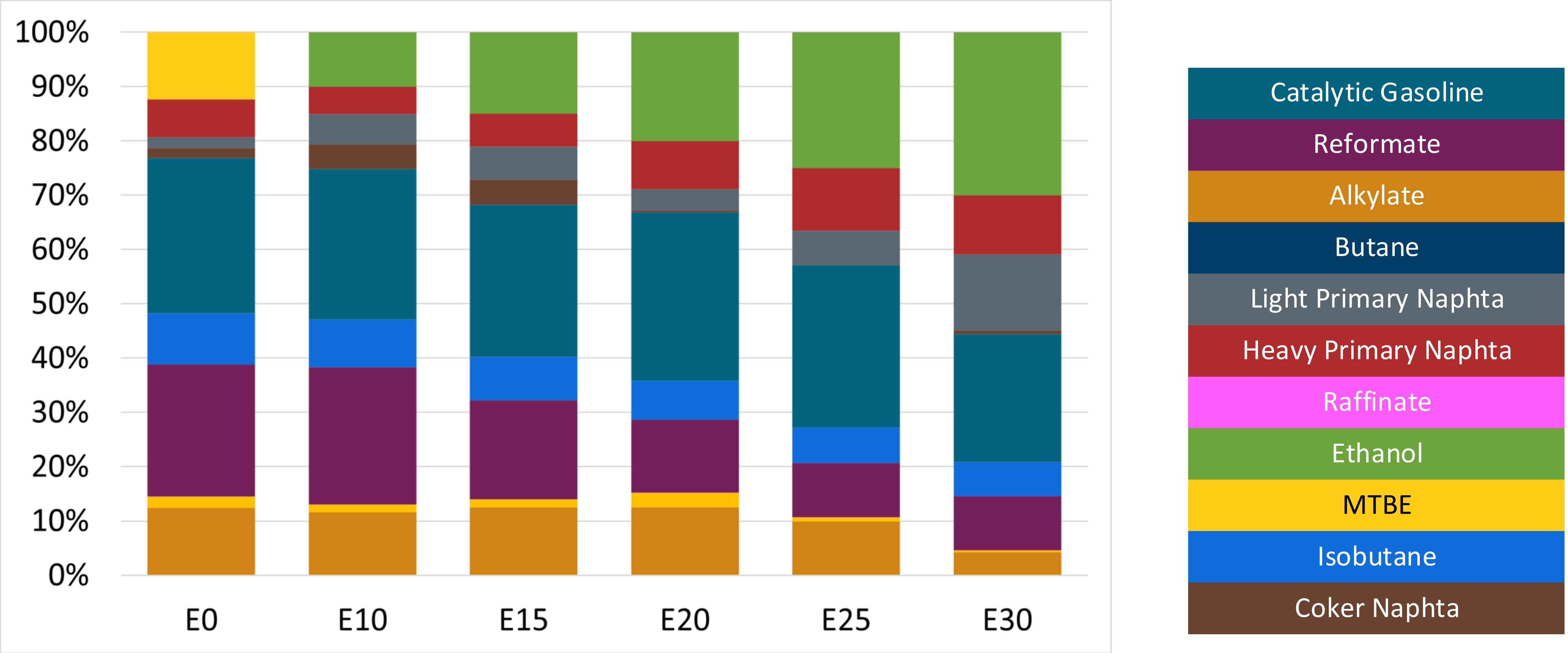
Source: Faro90

Regular Gasoline RP – Constant Octane



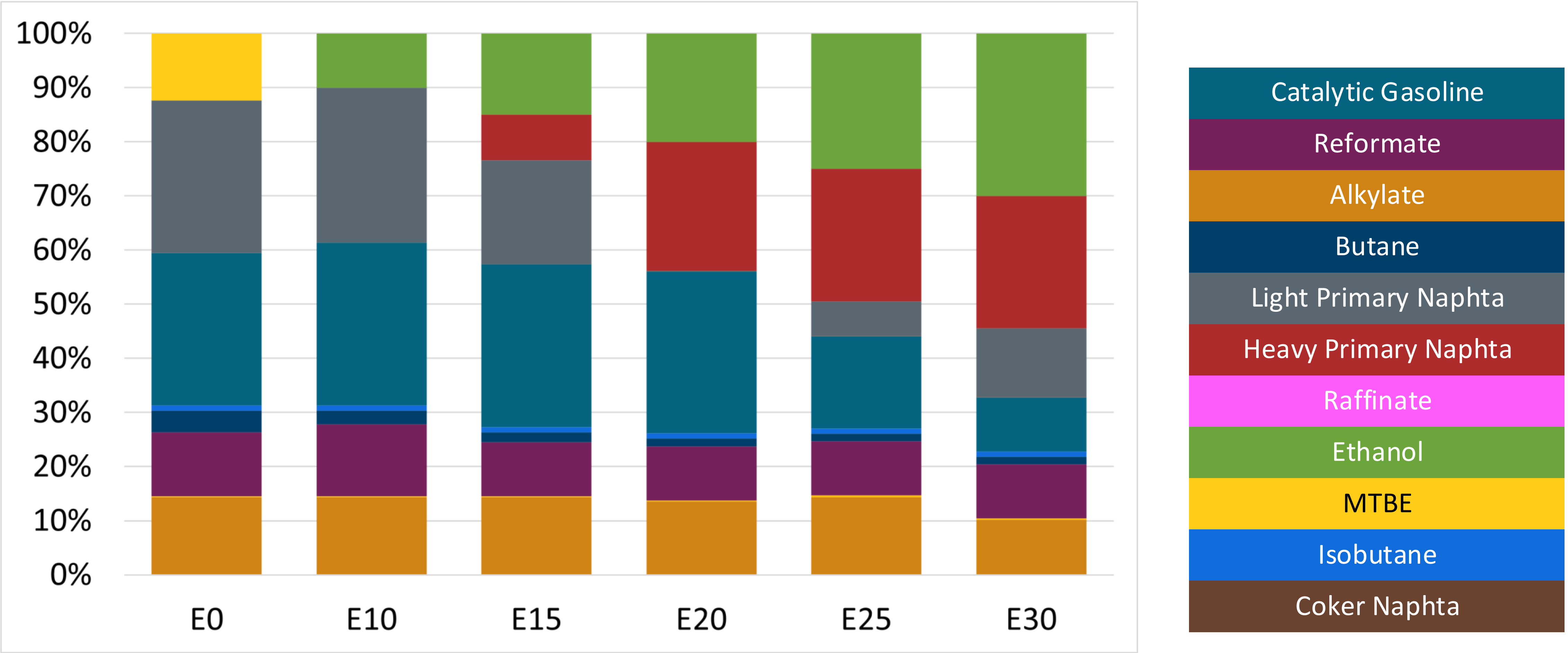
Octane (AKI)	87.1	87.0	87.0	87.0	87.0	87.0
Price (US$/gal)	$ 2.245	$ 2.286	$ 2.210	$ 2.138	$ 2.067	$ 2.005

Premium Gasoline RP – Constant Octane



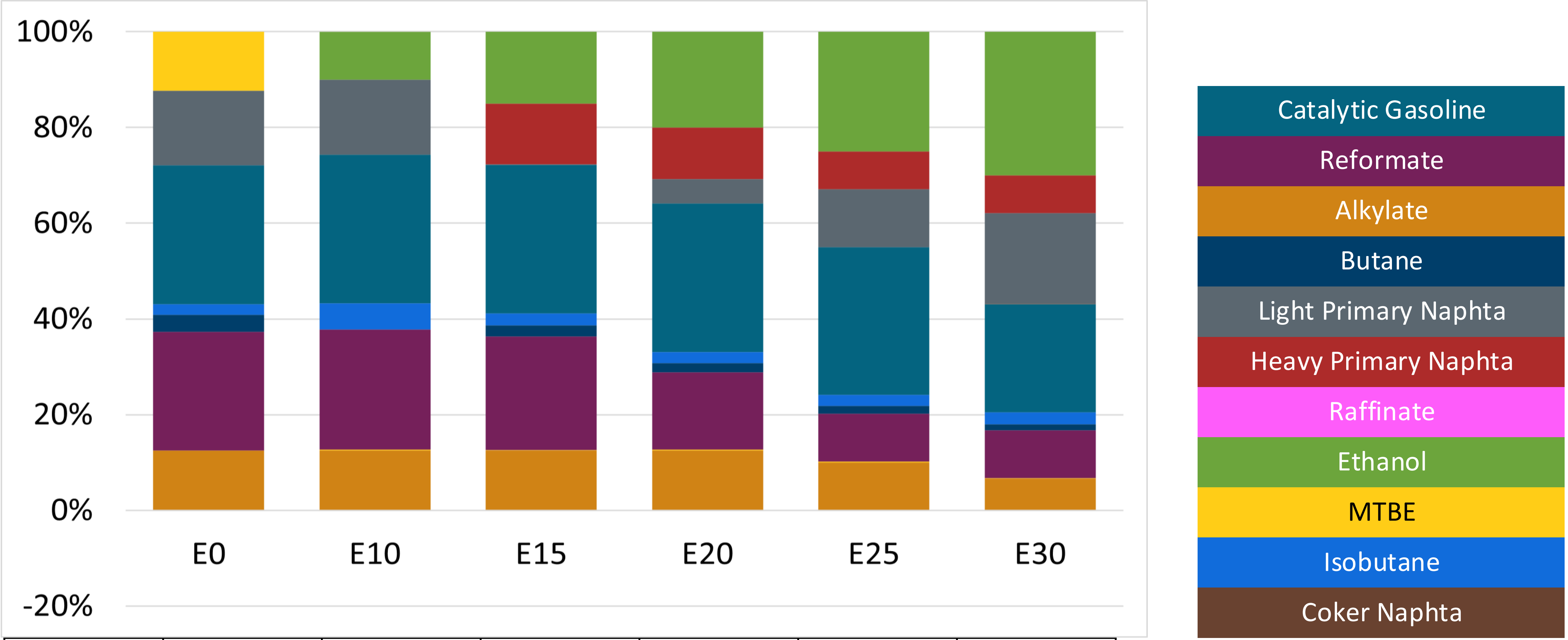
Octane (AKI)	91.0	91.0	91.0	91.0	91.0	91.0
Price (US$/gal)	$ 2.423	$ 2.395	$ 2.331	$ 2.266	$ 2.201	$ 2.134

Regular Gasoline ZM – Constant Octane



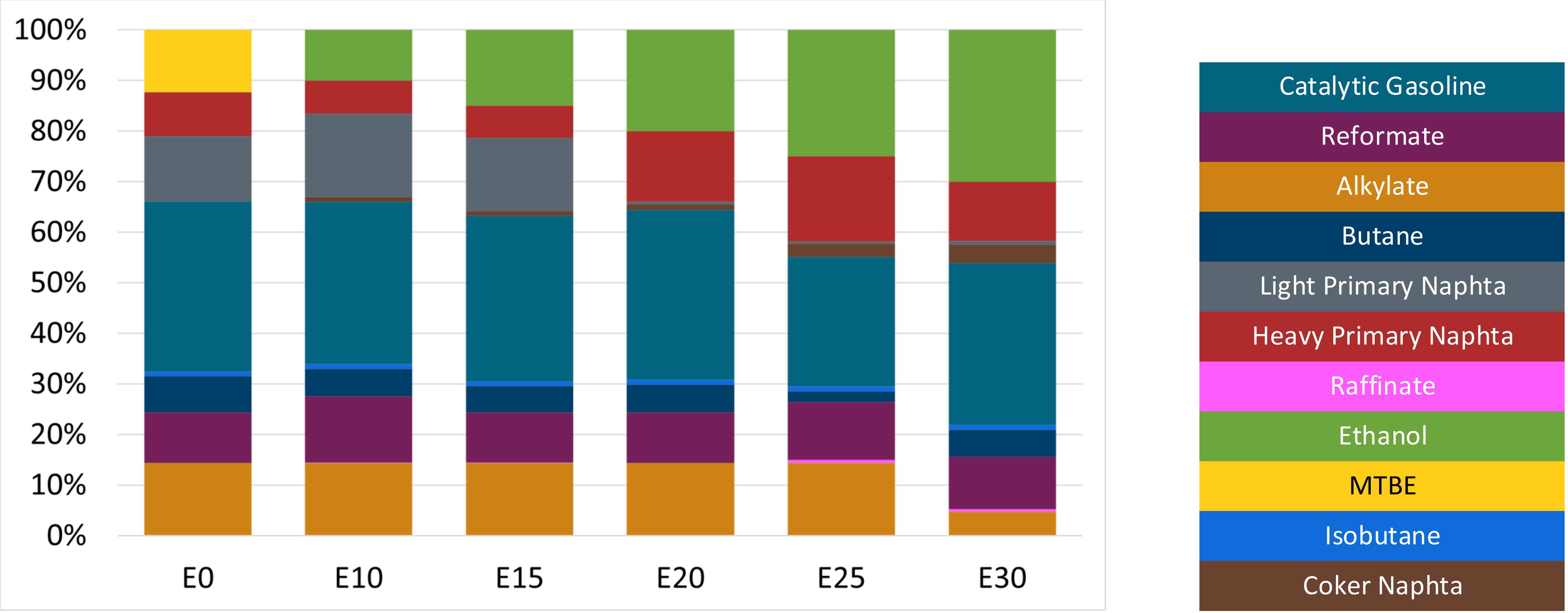
Octane (AKI)	87.0	87.0	87.0	87.0	87.0	87.0
Price (US$/gal)	$ 2.280	$ 2.265	$ 2.199	$ 2.131	$ 2.061	$ 1.990

Premium Gasoline ZM – Constant Octane



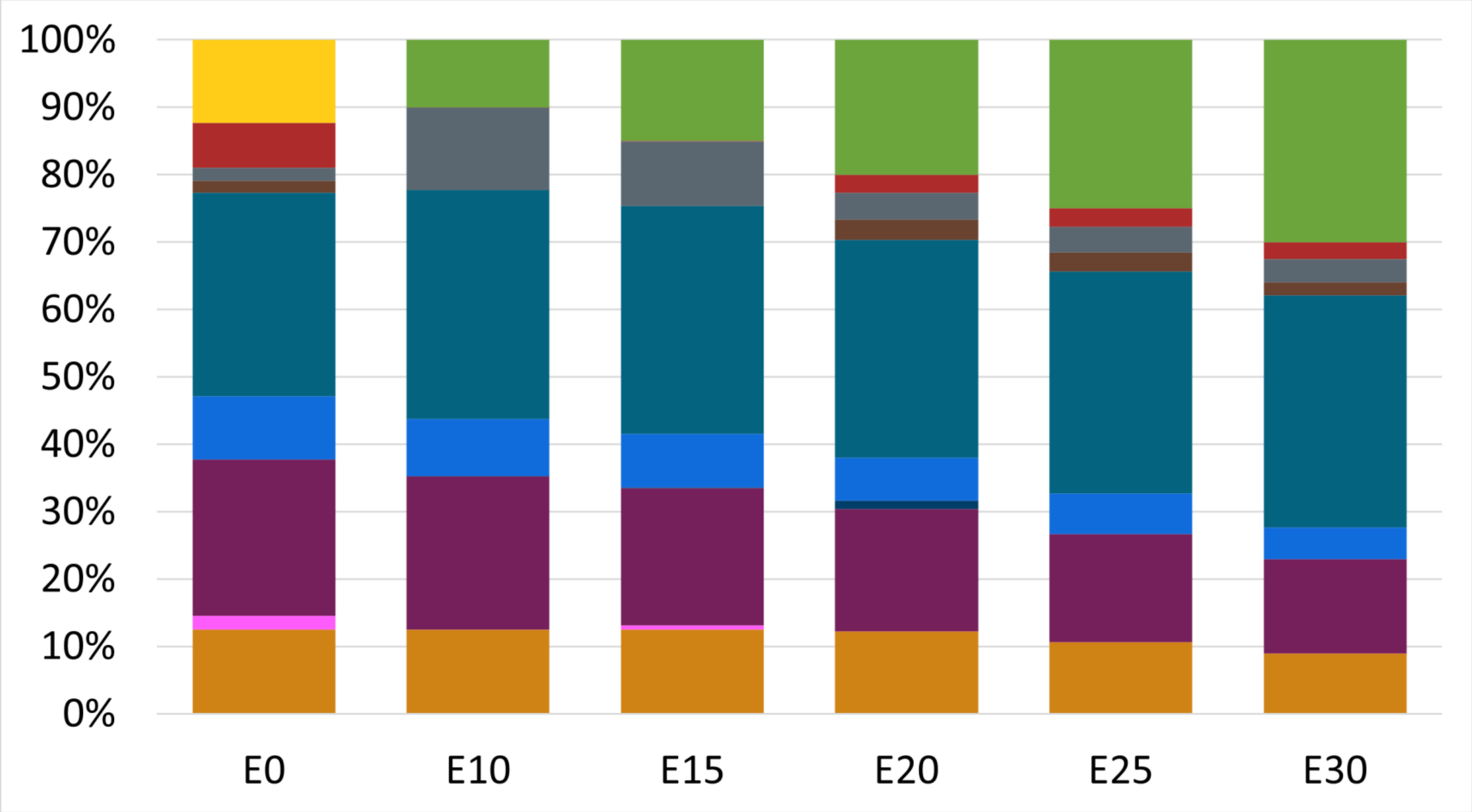
Octane (AKI)	91.0	91.0	91.0	91.0	91.0	91.0
Price (US$/gal)	$ 2.478	$ 2.442	$ 2.377	$ 2.310	$ 2.242	$ 2.172

Regular Gasoline RP – Increased Octane



Octane (AKI)	87.0	87.2	88.6	90.1	90.5	92.9
Price (US$/gal)	$ 2.423	$ 2.423	$ 2.423	$ 2.423	$ 2.423	$ 2.423

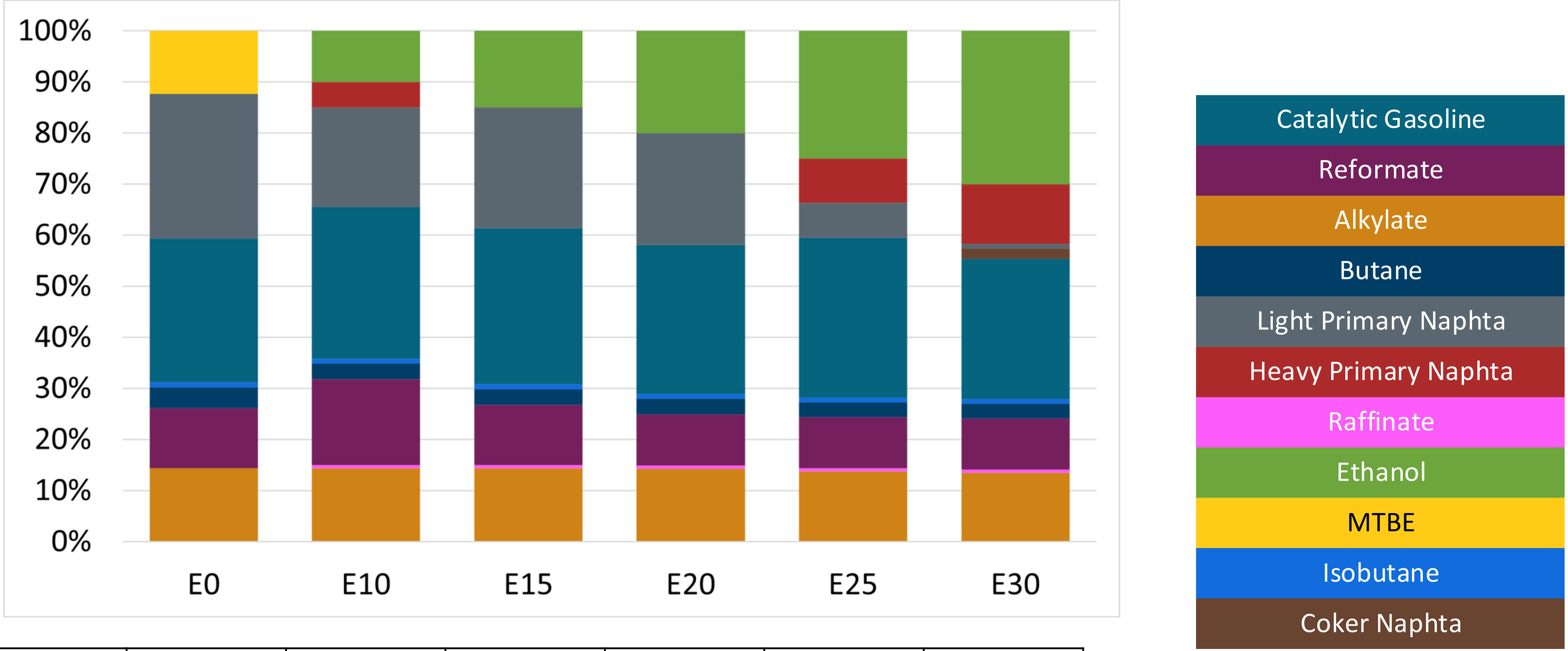
Premium Gasoline RP – Increased Octane



Octane (AKI)	91.0	91.6	92.8	94.1	95.1	96.2
Price (US$/gal)	$ 2.423	$ 2.423	$ 2.423	$ 2.423	$ 2.423	$ 2.423

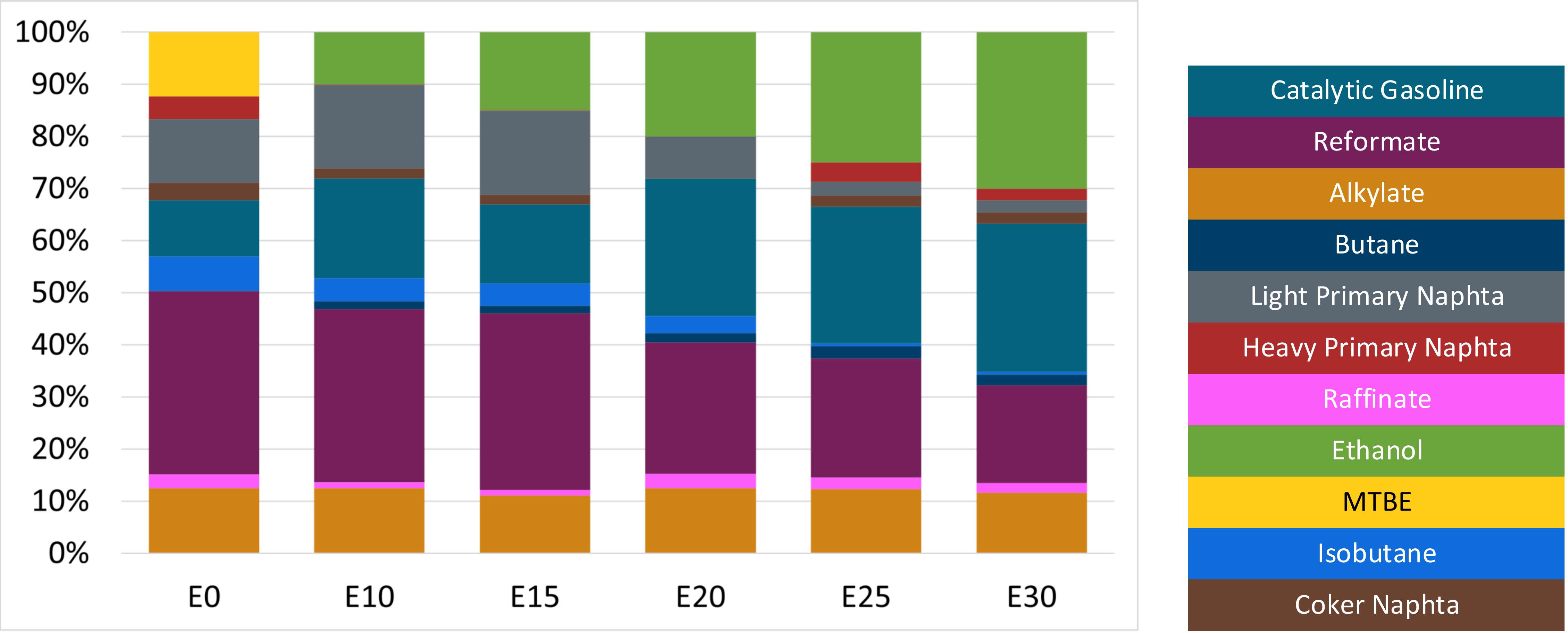
- Catalytic Gasoline
- Reformate
- Alkylate
- Butane
- Light Primary Naphta
- Heavy Primary Naphta
- Raffinate
- Ethanol
- MTBE
- Isobutane
- Coker Naphta

Regular Gasoline ZM – Increased Octane



Octane (AKI)	87.0	87.4	88.9	90.4	91.7	93.2
Price (US$/gal)	$ 2.280	$ 2.280	$ 2.280	$ 2.280	$ 2.280	$ 2.280

Premium Gasoline ZM – Increased Octane



Octane (AKI)	91.0	91.6	93.0	94.3	95.1	96.5
Price (US$/gal)	$ 2.475	$ 2.475	$ 2.475	$ 2.475	$ 2.475	$ 2.475

Vehicle Emission Impact for Ethanol Gasoline Blending

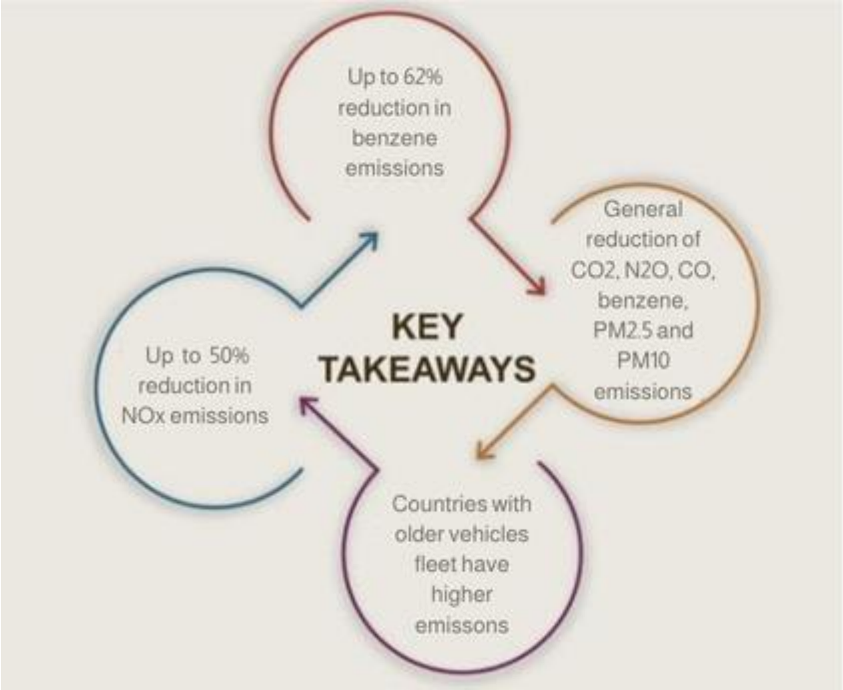
The model used in this analysis takes as a reference the **International Vehicle Emissions Model (IVE).**
The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

Main Results



**Source: Bureau of transportation statistics.*

Vehicular Fleet in Mexico

FARO90 HCX

Type	Heavy	Motor	Exhaust	Recuperator	Age	No. Vehículos
Auto/Sml Truck	Light	Carburator	3-Ways	PCV Tank	+8years	14.255
		Single FI	3-Ways		+8years	1.638.045
		Multi FI	3-Ways		+8years	6.559.661
			Euro III		+8years	3.064.588
			Euro IV		+8years	2.425.660
					4-8years	3.611.585
			Euro V		4-8years	855.809
					-4years	2.303.300
	Medium	Carburator	3-Ways	PCV Tank	+8years	2.024
		Single FI	3-Ways		+8years	64.686
		Multi FI	3-Ways		+8years	999.381
			Euro III		+8years	1.065.314
			Euro IV		+8years	770.022
					4-8years	1.274.081
			Euro V		4-8years	386.678
					-4years	1.374.040
Sml Engines	Light	4-Cycle	Euro II	PCV	+8years	2.714.667
					4-8years	1.945.800
			Euro III		-4years	2.797.515
Truck	Heavy	Carburator	3-Ways	PCV	+8years	19.293
		Fuel-Injection	3-Ways		+8years	2.132.687
			Euro III		+8years	1.222.113
			Euro IV		+8years	722.270
			Euro V		4-8years	904.413
					-4years	841.493

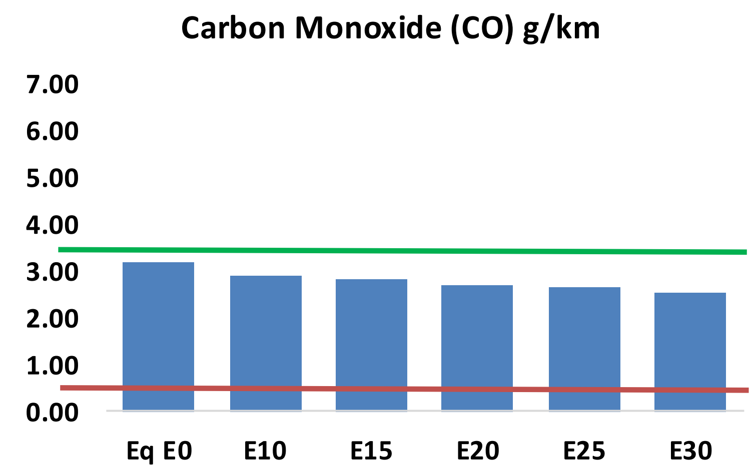
Total = 60,978,233

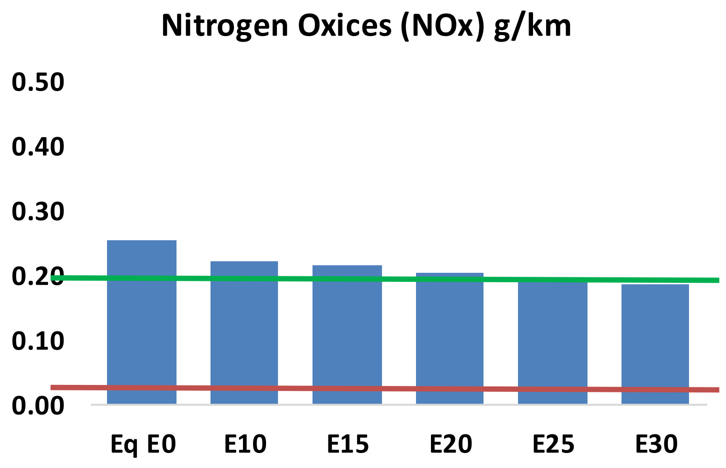
Gasoline Vehicular Fleet: **46,767,837**
Average Age: **10 years**

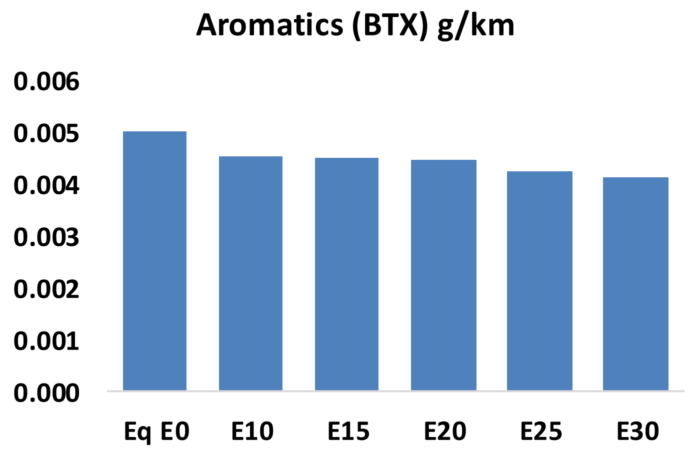
Source: INEGI

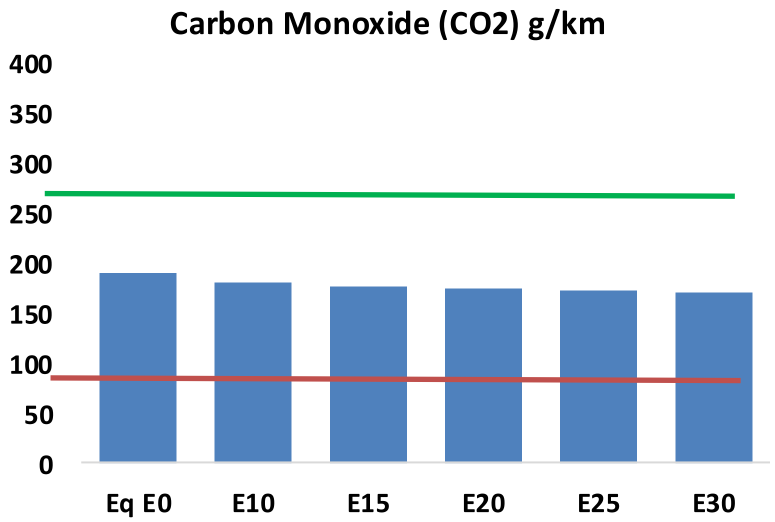
Vehicular Emissions in Mexico

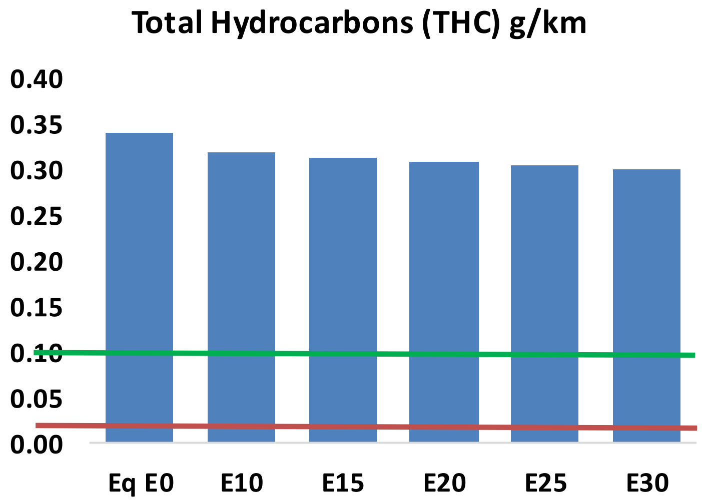
United States
Euro 6

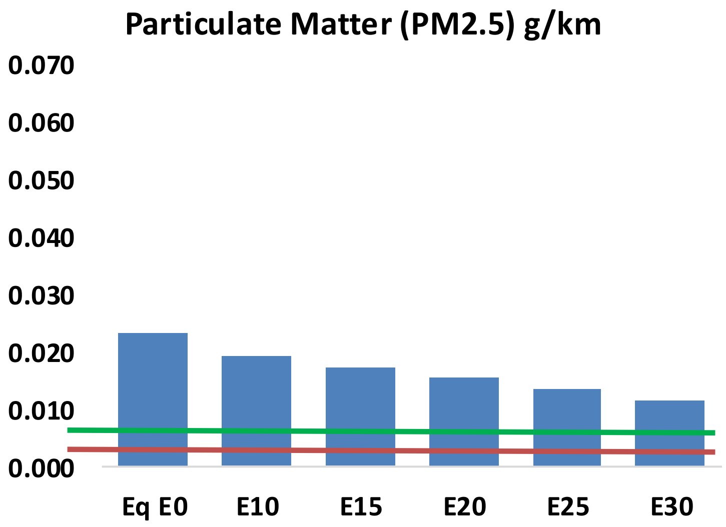












Source: Faro90, 2025.

Mexico – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	3,756,123	3,581,907	3,407,692	3,287,278	3,176,391	3,094,230	2,983,504	-9%	-15%
CO2	222,198,681	216,485,000	211,088,747	206,844,752	204,744,657	203,598,569	199,845,662	-5%	-8%
VOC	398,231	385,700	373,169	366,429	361,270	357,793	351,424	-6%	-9%
NOx	299,828	276,842	260,897	252,685	239,555	229,218	217,074	-13%	-20%
SOx	2,532	2,341	2,150	1,988	1,832	1,664	1,487	-15%	-28%
NH3	74,987	74,245	73,503	73,852	74,682	75,267	75,752	-2%	0%
N2O	3,036	3,019	3,007	3,022	3,084	3,118	3,154	-1%	2%
CH4	100,570	100,570	100,570	102,079	104,291	106,202	108,012	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	3,683	3,602	3,521	3,494	3,485	3,484	3,463	-4%	-5%
Acetaldehyde	8,141	9,343	13,709	20,877	28,443	32,502	38,405	68%	249%
Formaldehyde	26,810	26,065	30,385	35,678	37,378	41,351	45,015	13%	39%
Benzene	5,887	5,658	5,361	5,276	5,232	5,004	4,842	-9%	-11%
PM2.5	13,538	12,307	11,076	9,993	8,934	7,798	6,609	-18%	-34%
PM10	13,699	12,454	11,209	10,113	9,040	7,891	6,688	-18%	-34%

Source: Faro90

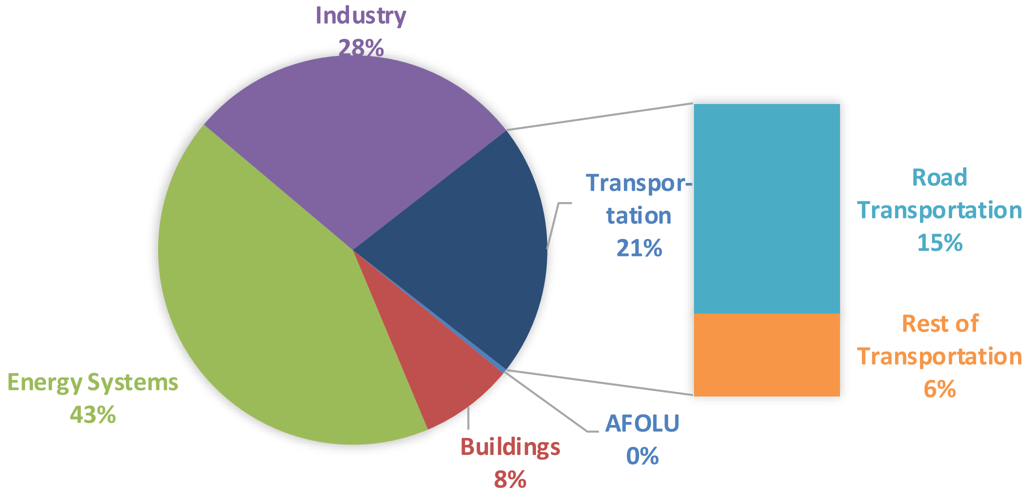
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

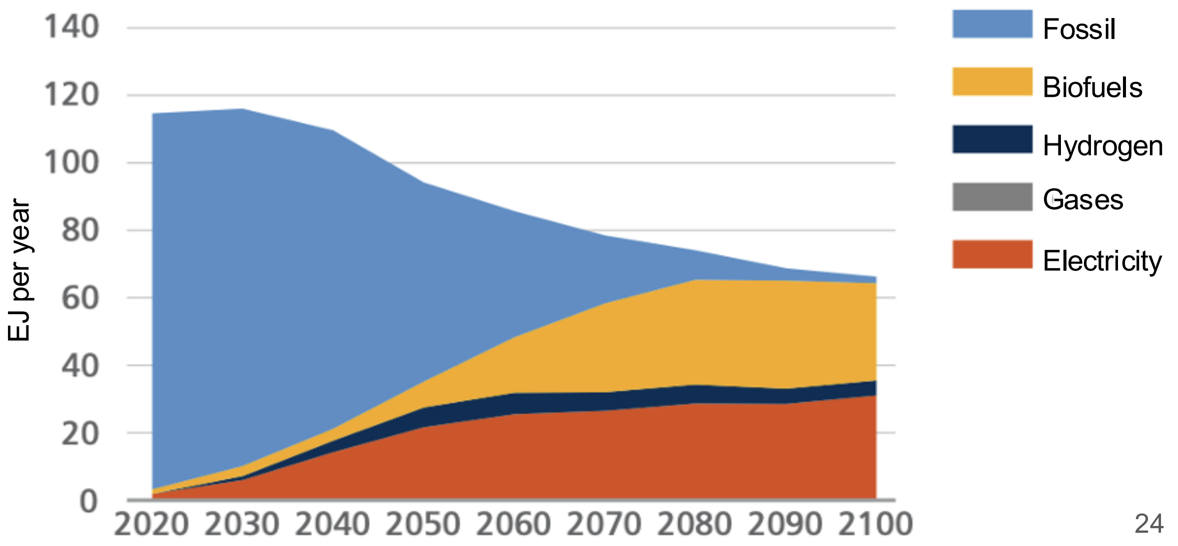
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE).** Results are compared to the current country targets for 2035. Current 2024 country emission are those published by **IPCC**.

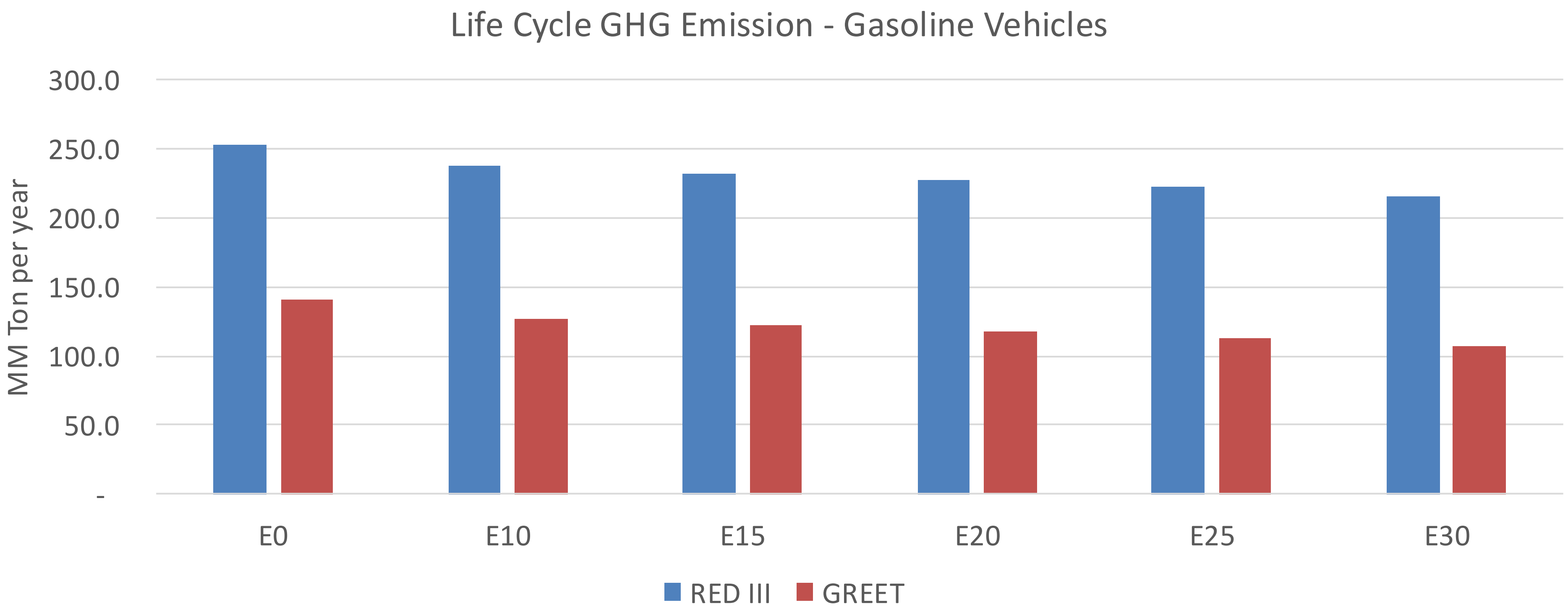
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Mexico – Gasoline Vehicle Fleet GEI Emissions



Country Emissions 2024 (MMT/year)	751.4
2035 Target (MMT/year)	500.4
Delta	251.0

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	9.4%	13.0%	16.5%	19.5%	23.5%
RED II/III	0.0%	5.9%	8.2%	10.3%	12.3%	14.6%

% 2035 Target	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	5.3%	7.3%	9.2%	10.9%	13.1%
RED II/III	0.0%	6.0%	8.2%	10.4%	12.4%	14.8%

Source: Greet, RED II/III, climateactiontracker.org, F90